

2023 Ch H1 Q11

Section: Nature's Chemistry

Topic: Proteins

Question Summary

When two amino acids react in a condensation reaction, water is eliminated and a peptide link is formed. Which of the following represents this process? (Options A-D show different structural diagrams)

Worked Solution

A peptide link is formed between the -COOH group of one amino acid and the -NH_2 group of another. During condensation, the -OH from the carboxyl group and an -H from the amine group combine to form water (H_2O). This leaves a -CO-NH- bond between the two residues.

Looking at the options:

- The correct diagram shows the elimination of water and the formation of a -CONH- linkage.
- The other options either show incorrect

bonding, missing atoms, or no peptide bond.

Final Answer

C — Correct peptide link formation

Revision Tips

- Proteins are formed by condensation polymerisation of amino acids.
- Each peptide bond forms with the loss of one molecule of water.
- The linkage is -CO-NH- , sometimes called an amide bond.